

HACK CORE 2026

PRAMAAN

Proof-driven, Multi-Agent AI for Biological Product Efficacy

- **Problem Track:** Track 03 - OUTCOME TRACKING & CONTINUOUS LEARNING
- **Problem Statement:** PS-05 - Season Journal - Field Outcome Logging for Biological Products
- **Team Name:** Unanimous 6
- **Team Leader:** Anushka Choudhari
- **Team Members:** 1) Shreya Gawade
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3) Tanisha Jagdale
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1. Executive Summary

Biological agricultural inputs like biofertilizers, biopesticides, and biocontrol agents show substantial potential for enhancing sustainable agriculture. Nevertheless, their efficacy in the field can vary widely based on the crop, location, soil type, weather conditions and farming practices employed. Currently, data on product usage at field level and subsequent results are siloed, resulting in weak regionally specific evidence for product efficacy. In essence, generic advice, opaque performance insights of products and poor farmer confidence in biological inputs.

PRAMAAN - Proof-driven Multi-Agent AI for Biological Product Efficacy, focuses on an offline-first mobile centric platform to translate field observations to a structured, evidence-backed agricultural intelligence. Farmers and field agents record a farmer can input product usage, observations of crop and photographs of product application and final yield results through a simple interface tailored for low-end Android devices and limited-connectivity conditions. A voice-first input from farmers help translate unstructured descriptions of the activities in the field into structured outputs.

The platform employs a multi-agent AI architecture to validate its field data, analyze crop photos, integrate climate and environment conditions, and detect outliers to generate seasonal efficacy reports. It leverages Firebase to achieve offline capture and sync and utilizes cloud based analytics to compile outcomes based on crop, location, product and season.

By employing these attributes, PRAMAAN establishes a **closed-loop learning system**: field usage -> observation -> results -> analysis -> evidence -> recommendations -> improved recommendations. As seasons progress and data accumulates PRAMAAN provides accurate and contextually relevant recommendations about a biological product.

Through its creation of a scalable real-world efficacy dataset, PRAMAAN benefits farmers, providing them with data-driven, informed decisions, industry stakeholders by providing them insights into real-world performance of biological products, retailers with credible product information and contributing structured agricultural data for AI farming practices.

2. Problem Understanding

Biological crop-protection products—including biopesticides, bio-fungicides, and bio-stimulants—can perform differently across India’s diverse agro-climatic conditions. However, there is currently no systematic, field-level feedback loop that consistently captures how these products perform under real-world farming conditions. This creates three critical gaps:

- **Recommendations remain generic** — Product and dosage recommendations are often not sufficiently tailored to local soil, weather, crop stage, and other field conditions.
- **Efficacy data remains limited** — Much of the available evidence is generated through controlled trials, while large-scale, real-world field outcomes remain insufficiently documented.
- **Farmer trust remains low** — Without accessible evidence demonstrating that a biological product has performed successfully under comparable farming conditions, farmers may remain hesitant to adopt biological solutions despite their potential contribution to sustainable and climate-resilient agriculture.

The challenge therefore calls for a lightweight, offline-capable, mobile-first solution that enables farmers and field agents to record biological product applications, in-season crop observations, and final yield outcomes. The collected data should then be aggregated and analyzed to generate automated, region-specific and crop-specific efficacy reports, creating a continuous feedback loop between field outcomes and future product recommendations.

3. Proposed Idea

“**Pramaan**” (प्रमाण) means *proof* or *evidence* in Hindi and Sanskrit. The name is deliberately chosen because the core deliverable of PS-05 is not simply a field-logging application, but a system that builds trustworthy, real-world evidence of biological product efficacy at scale.

PRAMAAN — Proof-driven Multi-Agent AI for Biological Product Efficacy — is a multi-agent AI system designed to operate behind a simple, voice-first, offline-capable mobile application. A supervisor agent coordinates specialised AI agents responsible for tasks such as speech understanding, crop and product image analysis, weather enrichment, data validation, efficacy analytics, and report generation.

The key challenge in PS-05 is not merely collecting data. Field information can be multilingual, unstructured, incomplete, and collected under limited-connectivity conditions. PRAMAAN therefore focuses on transforming this messy field input into a structured, validated, and analysable dataset that can generate reliable efficacy insights.

3.1 Why an Agentic Architecture?

A single monolithic AI call would have to handle several fundamentally different tasks, including speech understanding, image analysis, environmental-context enrichment, anomaly detection, analytics, and report generation. These tasks have different objectives and potential failure modes. PRAMAAN instead assigns specialised responsibilities to individual agents, making each component easier to evaluate, improve, and maintain independently.

As real-world field data grows, individual agents can be refined or replaced without redesigning the entire system. This modular architecture also provides a foundation for integrating additional agricultural intelligence services as the platform scales.

PRAMAAN is designed around Google's agentic ecosystem and can use the **Agent2Agent (A2A) protocol** to enable standardized communication between compatible AI agents. This creates the potential to connect PRAMAAN with other agricultural AI services, including multilingual farmer-assistance and weather-intelligence systems, as compatible integrations become available.

4. Objectives

The overarching goal of PRAMAAN is **to implement a closed-loop, field-level outcome tracking system** that captures, translates, and aggregates real-world product outcomes into reliable and actionable efficacy intelligence for individual regions and crops.

Specifically, the objectives of PRAMAAN include:

1. **Deliver effortless field-level data capture:** Providing farmers and field agents a user-friendly, intuitive mobile interface for collecting biological product application data, field observations and the final yield outcomes using minimum effort.
2. **Ensure performance in low-connectivity environments:** Facilitating seamless and efficient data collection from feature-rich and low-end Android devices irrespective of internet connectivity, ensuring automatic data synchronization once connection is available.
3. **Structure un-structured field inputs into reliable agricultural data:** Utilizing voice-first multilingual AI, intelligent input methods, images and document scanning to effectively convert natural language observations from farmers into structured data inputs.
4. **Enhance data quality and reliability:** Automatically validate collected field data to avoid anomalies, duplicates and also to append crucial context to records for added reliability and credibility.
5. **Generate localized biological product efficacy insights:** Collating field data related to product application, on-field observations, climatic factors, and eventual yields to rigorously evaluate product effectiveness across diverse crops, geography and time periods.
6. **Provide automated seasonal efficacy summaries:** Automatically generate succinct, yet comprehensively evidence-backed seasonal reports summarizing performance findings and related confidence levels for diverse audiences.
7. **Foster a closed-loop, continuously learning system:** Leveraging the cumulatively acquired real-world outcome data to progressively improve and inform better recommendations for biological products over time.
8. **Develop an accessible and evidence-based agricultural dataset:** Constructing and making accessible a well-structured and validated data set containing real-world biological product outcome data to enable better evidence-based decisions among various agriculture stakeholders.

5. User Workflow

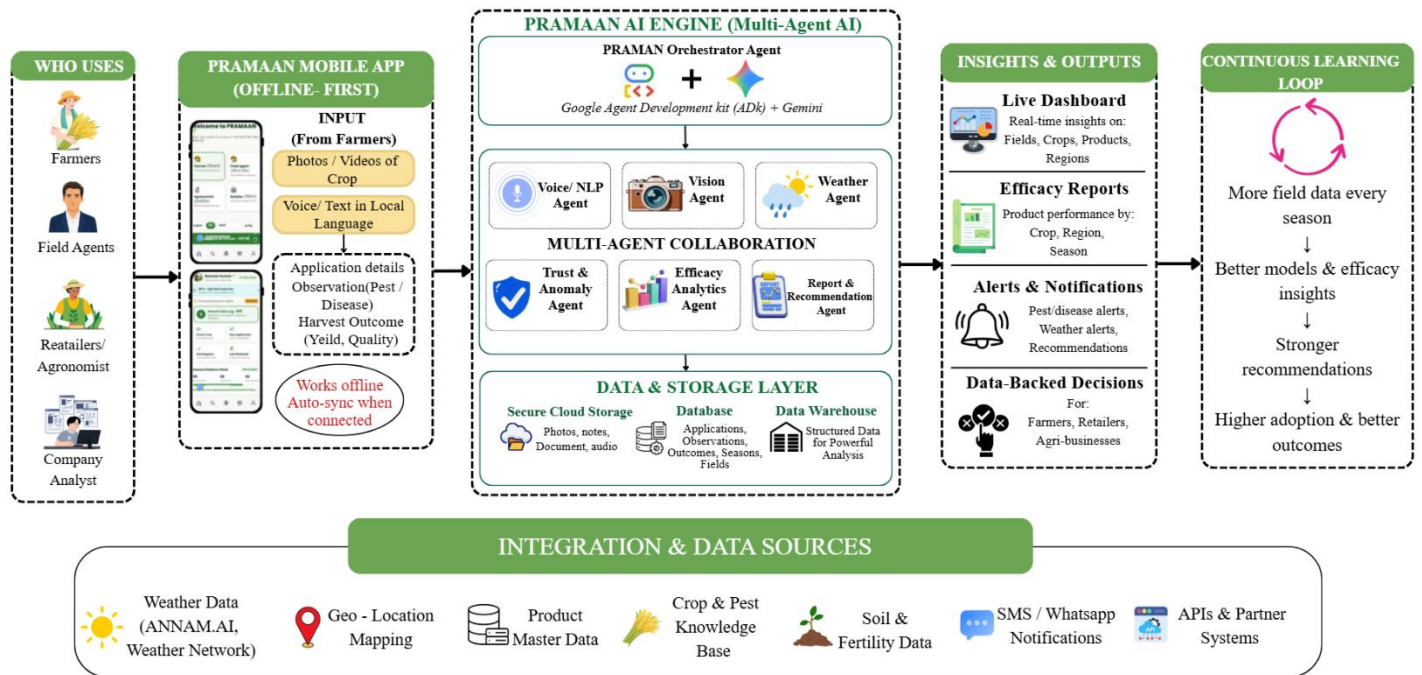


Fig. 1. Pramaan – Field Outcome Tracking Workflow

6. The Orchestrator Agent

PRAMAAN uses a modular multi-agent AI architecture to transform raw field inputs into validated, contextualized, and actionable efficacy insights. A central **Orchestrator Agent** coordinates specialised agents, each responsible for a specific task such as voice processing, image analysis, weather enrichment, data validation, efficacy analysis, and report generation.

This architecture separates complex agricultural intelligence tasks into smaller, independently testable components. As real-world field data grows, individual agents can be refined or replaced without redesigning the complete system.

6.1 The Orchestrator Agent

The **Orchestrator Agent** acts as the central coordinator of PRAMAAN. It receives structured field events from the mobile application and determines which specialised agent should process each task.

It coordinates the outputs of the Voice/NLP, Vision, Weather, Trust & Anomaly Detection, Efficacy Analytics, and Report Generation agents, ensuring that information from different sources is combined into a consistent field-level record and downstream efficacy insight.

The orchestrator enables a modular architecture in which individual agents can be independently evaluated, improved, or replaced as more real-world field data becomes available.

PRAMAAN follows a modular multi-agent architecture in which individual agents handle specific field-data and decision-support tasks. For the Build Sprint, the team will prioritize the core agents required to demonstrate the end-to-end PS-05 workflow, while the remaining agents represent extensible capabilities of the proposed architecture.

6.2 Specialised AI Agents

Agent	Core AI Technique	Responsibility
Voice & Text Agent	Gemini	Converts farmer's voice into structured information such as crop, product, dose, stage and field observations.
Vision Agent	Gemini multimodal (vision)	Analyzes uploaded field/product images, helps identify relevant

		information and detects unclear or duplicate images.
Weather & Environmental Context Agent	Meteoblue + CE Hub APIs	Adds historical and forecast weather information to the field record to understand how environmental conditions affect product performance.
Trust & Anomaly Detection Agent	Gemini + basic validation rules	Checks field records for missing, inconsistent, duplicate or unusual information before analysis.
Efficacy Analytics Agent	Vertex AI + BigQuery ML	Compares product performance across crops, regions and seasons to identify useful efficacy patterns.
Report Generation Agent	Gemini	Converts the analyzed results into simple, evidence-based reports and answers questions about product performance.

7. Data & Analytics

The problem statement explicitly needs adoption by farmers who may not be comfortable typing structured forms. PRAMAAN treats natural-language, voice-first interaction as the primary interface, not an add-on:

1. Farmer speaks a sentence such as: “I sprayed Product X on my cotton field two days ago, half a litre per acre, plants look healthy.”
2. The Voice/NLP Intake Agent transcribes it (multilingual ASR), then runs entity extraction to populate: product, crop, quantity, unit, application date, and a free-text observation field.
3. If a field is missing or ambiguous, the orchestrator agent generates one targeted follow-up question in the same language — instead of showing a blank form — keeping the interaction conversational.
4. The same NLP pipeline powers the query side: agronomists or Syngenta analysts can ask the Report Generation Agent questions in plain English/Hindi and get grounded answers pulled from BigQuery, not hallucinated ones.

7.1 Dataset & Organizer Resources

Primary organizer-provided resources used by PRAMAAN:

Resource	Intended Use
Meteoblue Historical Dataset API	Retrieve historical weather conditions associated with field location and application/observation periods for environmental-context and efficacy analysis.
CE Hub Forecast API	Provide forecast information that can support field-level environmental context and decision-making.
Organizer-provided Algorithm Logic	Incorporate the prescribed algorithmic logic into relevant analysis/recommendation workflows.

PRAMAAN primarily relies on the resources provided by the organizers to maintain alignment with the challenge evaluation criteria. No additional proprietary or non-organizer datasets are required for the core Build Sprint prototype.

8. Technology Stack

Layer	Technology	Why
Mobile client	Flutter (Android-first, low-end device support)	Single codebase, small APK size, works on entry-level Android phones per the challenge's device constraint
Offline data & sync	Firebase (Firestore offline persistence, Auth, Cloud Messaging, Cloud Functions)	Matches the challenge's own "Technology Angle" guidance; mature offline-first primitives
Agent orchestration	Google Agent Development Kit (ADK) + Agent2Agent (A2A) protocol	Code-first, model-agnostic multi-agent orchestration; deployable to Vertex AI Agent Engine; A2A lets PRAMAAN's agents interoperate with other ANNAM.AI / Syngenta agents
Foundation models	Gemini (multimodal: speech, vision, text) via Vertex AI Model Garden	Single model family covers ASR, NLU, vision QC and grounded report generation
Analytics	BigQuery + BigQuery ML / Vertex AI for trend and efficacy modelling	Scales from a few hundred pilot logs to a national dataset without re-architecture
Storage	Cloud Storage (images/audio) with Firestore metadata references	Keeps heavy media out of the hot database path
Deployment / DevOps	Vertex AI Agent Engine (managed runtime), Cloud Run for lightweight services	Autoscaling, session management, minimal ops overhead for a student team

9. Innovation & Differentiation

PRAMAAN goes beyond conventional agricultural data-collection and advisory applications by creating a continuous evidence-generation loop between field-level biological product use and future recommendations. Its differentiation lies not in a single AI feature, but in the integration of **offline-first field logging, voice-first interaction, multi-agent intelligence, contextual enrichment, efficacy analytics, and continuous learning** within one system.

9.1 What Makes PRAMAAN Different?

Existing approach	Gap	How PRAMAAN differs
Generic crop-health apps (e.g., Plantix-style diagnosis apps)	Diagnose crop problems but don't close the loop on which treatment actually worked	PRAMAAN is outcome-tracking, not just diagnosis — it links application → observation → yield
Enterprise agri-data platforms (e.g., Cropin's OrbitAI on Google Cloud)	Built for enterprise-wide farm operations, not lightweight farmer-level biological efficacy logging	PRAMAAN is purpose-built, lightweight, and specifically closes the biological-efficacy feedback loop the problem statement names

9.2 Core Innovation

The central innovation of PRAMAAN is its ability to transform individual field events into cumulative evidence:

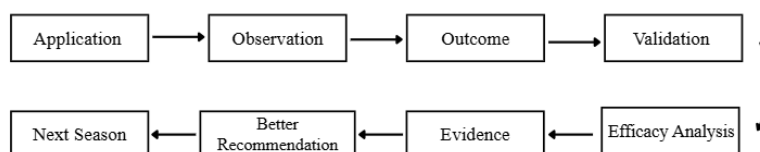


Fig. 2. Praman Stages

Instead of treating each field record as an isolated data point, PRAMAAN connects records across **product, crop, region, and season** to progressively build a real-world evidence base for biological products.

This creates a shift from **data collection to evidence generation**, and from **one-time recommendations to continuous learning**.

10. Feasibility & Implementation Plan

PRAMAAN is designed as a modular, cloud-ready system that can be developed incrementally, beginning with a lightweight field-logging MVP and progressively adding advanced AI capabilities. The proposed technology stack uses established frameworks and managed cloud services, reducing infrastructure complexity while allowing the solution to scale from a pilot deployment to national-level field data collection.

10.1 Build Sprint MVP Boundary

In order to achieve practical implementation within the scope of the Build Sprint, the PRAMAAN project will focus on an end-to-end MVP functionality rather than trying to implement all of the long-term architectural aspects.

PRAMAAN Build Sprint MVP will showcase the whole core flow:

Offline Field Logging -> Voice-Assisted Structured Capture -> AI Validation -> Organizer-Provided Environmental Enrichment -> Cloud Sync -> Basic Efficacy Analytics -> Evidence-Based Report

The main pieces to be developed within the Build Sprint are the following:

1. Flutter mobile application to log biological product applications, crop observation, and results.
2. Offline-first data logging and synchronization with Firebase.
3. Voice-assisted field logging with the use of Gemini service that converts farmer observations into structured fields.
4. AI validation to detect any missing, contradictory, duplicated or otherwise implausible records.
5. Organizer-provided integration of weather/resource conditions using available Meteoblue and CE Hub services for environmental context.
6. Basic product x crop x region efficacy analysis with use of logged field outcomes.
7. Evidence-based seasonal report with use of structured analytics results.

The advanced features like multi-agent coordination, A2A interoperability, vision analysis, big machine learning and continuous learning will represent the scalable architecture for future developments and will not

10.2 Phased Implementation

Phase 1 — Field Data Capture

Develop the Flutter application, field/season registration, application logging, observation logging, outcome recording, and offline synchronization.

Phase 2 — AI Intelligence

Integrate the Voice/NLP, Vision, Weather, and Trust & Anomaly agents to enrich and validate field records.

Phase 3 — Efficacy Analytics

Introduce product–crop–region analysis, outcome comparisons, trend detection, and automated seasonal efficacy reports.

Phase 4 — Continuous Learning & Scale

As the volume of verified field outcomes increases, apply advanced analytics and machine learning to identify recurring patterns and improve region- and crop-specific recommendations across subsequent seasons.

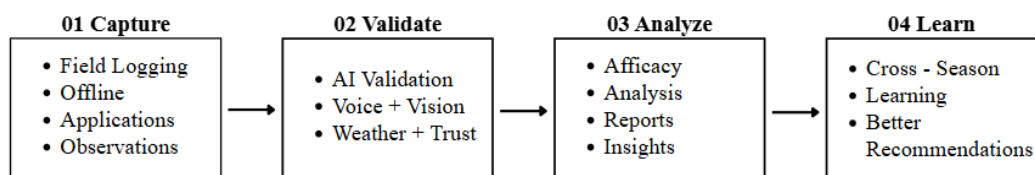


Fig. 3. Implementation Phases

10.3 Scalability & Deployment

The modular architecture allows individual components to scale independently. Firebase provides the initial backend and synchronization layer, while Cloud Storage supports scalable media storage. BigQuery and Vertex AI can be introduced as data volume grows for large-scale analytics and machine-learning workloads. This approach allows PRAMAAN to begin with a focused, demonstrable MVP while maintaining a clear technical path toward a national-scale biological product outcome intelligence platform.

11. GitHub Repository & Development Status

GitHub

Repository:

[https://github.com/Rohit22o9/pramaan_hackcore_2026.git]

11.1 Team Contribution

Team Member	Expected Role / Contribution
Anushka Choudhari	Team coordination, Flutter mobile application, AI integration
Shreya Gawade	Data analytics, efficacy analysis, dashboards and cloud integration and overall system architecture
Shravani Misal	AI/ML pipeline, Gemini integration and AI-agent development
Tanisha Jagdale	UI/UX design, field workflow and application testing
Rohit Ratnaparkhi	Backend development, Firebase integration and data management

12. Expected Impact

PRAMAAN creates value across the biological-product ecosystem by transforming field-level experiences into structured, trustworthy evidence. Its impact extends beyond individual farm records to farmers, product teams, retailers and distributors, and the broader agricultural AI ecosystem.

➤ **Farmers**

PRAMAAN provides documented evidence of what worked under field conditions similar to their own, helping reduce dependence on generic product advice. Over time, accumulated field outcomes can support more personalised, crop- and region-specific recommendations for subsequent seasons.

➤ **Syngenta**

PRAMAAN enables the creation of a proprietary, continuously growing real-world efficacy dataset that complements controlled trials with evidence from diverse field conditions. This dataset can help identify product-performance patterns across crops, regions, seasons, and environmental conditions.

➤ **Retailers & Distributors**

PRAMAAN provides evidence-backed information that can support biological product recommendations at the point of sale. By connecting recommendations with documented field outcomes, retailers and distributors can help build greater farmer confidence and trust in biological products.

➤ **ANNAM.AI / National Agri-AI Ecosystem**

PRAMAAN contributes structured, geo-tagged, and weather-correlated field outcome data to the broader agricultural-intelligence ecosystem. This creates opportunities to connect real-world biological product performance with existing agricultural AI capabilities, including weather intelligence and multilingual farmer interaction systems.

13. References

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14. Conclusion

PRAMAAN is designed to close the critical feedback gap between **biological product application and real-world field outcomes**. Instead of treating field logging as an end in itself, PRAMAAN transforms applications, crop observations, environmental conditions, and final outcomes into a structured and continuously growing evidence base.

Its **offline-first, voice-enabled mobile experience** makes field-level data capture practical even in low-connectivity environments, while its **multi-agent AI architecture** enables the collected information to be validated, enriched, analyzed, and converted into actionable efficacy insights. By connecting product, crop, region, season, and outcome data, PRAMAAN moves beyond generic recommendations toward evidence-based agricultural intelligence.

The core value of PRAMAAN is its **closed-loop learning model**:

Capture → Validate → Analyze → Learn → Recommend → Improve

With every verified field outcome, the system becomes better informed about how biological products perform under diverse real-world conditions. This creates value not only for farmers, but also for retailers and distributors, product teams, and the broader agricultural AI ecosystem.

Ultimately, PRAMAAN aims to make every season a source of learning—building the evidence needed to strengthen farmer trust, improve biological-product recommendations, and enable a more data-driven and sustainable future for Indian agriculture.